Wave Aspect: Electron creates field; field acts on another electron.
Particle Aspect: Electron emits photon; photon is absorbed by another electron.

Both statements describe the same event. The first version is familiar to everybody who has ever had any dealings with electric fields. The second one is unfamiliar to most people, because in technical science as well as in macroscopic physics it is always unnecessary to conceive of an electric field as linked to photons. Under atomic conditions, however, this very frequently proves to be a useful expedient. With reference to atomic radiation, it is often more convenient to speak of photons than of spherical waves.

Now let us apply exactly the same type of phraseology to the forces operative between protons and neutrons. First, we can say: The neutron produces a nuclear field, and this field acts on the proton. This is the description in terms of the wave aspect. In the terminology of the particle aspect, our description will be: The neutron produces particles, and these particles are absorbed by the proton. Let us again express this schematically, as follows:

Wave Aspect: Neutron creates field; field acts on proton.
Particle Aspect: Neutron emits electron plus neutrino; electron and neutrino are absorbed by proton.

Interpreting in this manner the force operative between neutron and proton, we see that an exchange of charge is linked with the action of the force. Namely, if in order to exert this force, the neutron must emit an electron and a neutrino, its charge is altered; it changes into a proton. And conversely, a proton changes into a neutron, due to the absorption of an electron and a neutrino. An exactly analogous conversion may occur also when a proton emits a positron and a neutrino, which are then absorbed by the neutron.

Thus, the nuclear forces are associated with an exchange of charged particles, and for this reason, forces of this kind are called *exchange forces.* They are of a very peculiar character, and it is their characteristic feature that their action is linked with an exchange of roles between the two partners. In this respect, therefore, they are totally different from the electric forces. But a close relationship with chemical forces is again evident. Quantum theory has